
WHITE PAPER

PetPulse

Egypt's all-in-one pet care ecosystem connecting owners, vets, trainers, and shops on a single trusted platform.

The Problem

Egypt's pet care sector is fragmented and largely offline — pet owners struggle to find verified veterinarians, certified trainers, or trusted pet sitters, while pet businesses lack a unified digital channel to reach customers, resulting in poor service discovery, unverified providers, and zero community infrastructure for Egypt's rapidly growing pet-owning population.

Our Solution

PetPulse is a full-stack web platform and marketplace that connects Egyptian pet owners with verified veterinarians, certified dog trainers, premium pet shops, adoption hubs, mating match services, monthly subscription boxes, and secure pet hosting — all in one place, with profile verification and community-driven trust signals.

Proof It Works

PetPulse's MVP launched on Vercel and successfully integrated 7 distinct service verticals — veterinary consultations, training academies, pet shops, adoption, mating, subscription boxes, and pet hosting — into a single responsive platform serving Cairo's pet community.

What We're Looking For

Partner with PetPulse to onboard your veterinary clinic, pet shop, or training academy onto Egypt's first unified pet care marketplace — reach thousands of verified pet owners from day one.

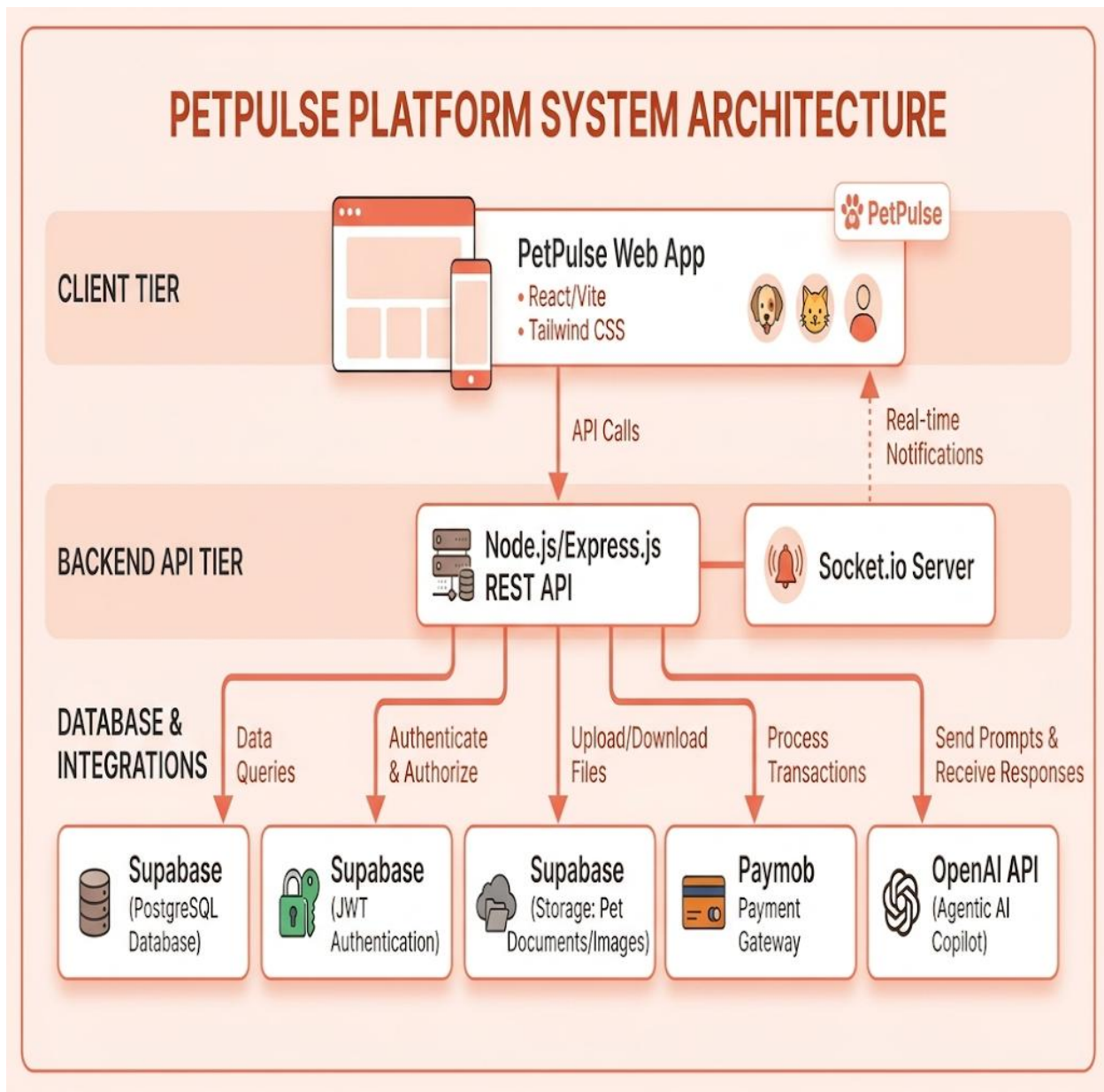
How It Compares

Feature	Our Solution	Facebook Groups	Generic Classifieds (OLX)
<i>Provider Verification</i>	✓ <i>Verified Profiles</i>	<i>Unverified</i>	<i>Unverified</i>
<i>Service Verticals</i>	✓ <i>7 (Vet, Train, Shop, Adopt, Mate, Box, Host)</i>	<i>1 (Community only)</i>	<i>1 (Listings only)</i>
<i>Pet-Specific UX</i>	<i>Yes — Purpose-built</i>	<i>No — Generic UI</i>	
<i>Egypt-Focused</i>	✓ <i>Yes</i>	<i>No</i>	<i>Partial</i>

Example values shown — replace with your real comparison.

2. How It Works

System at a Glance



The 4-Step Process

1	Sign Up & Build Your Profile	<i>Pet owners, vets, trainers, and shop owners create verified profiles — specifying their pets, services, and availability. Businesses upload credentials for trust-badge verification.</i>
2	Browse & Connect	<i>Owners search by service type, location, and pet breed. PetPulse surfaces verified vets, trainers, shops, adoption listings, and hosting providers near them in Cairo.</i>
3	Book & Transact	<i>Users schedule appointments, place product orders, submit hosting requests, or contact providers directly through the platform — eliminating the need for scattered WhatsApp conversations and unreliable service discovery.</i>
4	Review & Grow	<i>After every interaction, users can leave verified reviews and ratings. These reviews help providers build credibility while helping other pet owners make informed decisions.</i>

3.Results & Validation

7 SERVICE VERTICALS	Cairo LAUNCH CITY	1 Platform ALL-IN-ONE SOLUTION
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Example KPI values shown — replace numerator, denominator, and labels with your real metrics.

4. Get Involved

Let's talk. We are looking for industry partners, investors, and collaborators to take this solution from prototype to deployment.		
FOR INDUSTRY Partner With Us <i>List your clinic, shop, or academy on PetPulse and gain instant access to CairoRun a real-world pilot at your facility with full</i>	FOR FUNDERS Invest in PetPulse <i>Help us scale from prototype to commercial deployment — detailed financials available on request.</i>	FOR RESEARCHERS Research & Expand <i>Partner with us on pet health data research, AI-driven vet triage, or regional expansion across MENA.</i>

support from our teams verified pet owner community.

CONTACT THE TEAM

Team members	Salma Mohamed – 230105870, Laila Mohamed – 250103991, Fatma Mohamed – 230106003, Sara Salah – 230105798, Ahmed Sherif – 230104417, Essam Osama – 230104858, Gaber Hosny – 230101439, Mohamed Ebrahim – 230101269, Mahmoud Mustafa Mahmoud – 230105193, Abdelrahman Essam Ali – 230102357
Supervisor	<i>Dr. Hend Adel · Email hend.ahmed@sut.edu.eg</i>
Project Page / Repo Qr Code for QMS	https://petpulse-pi.vercel.app/

PetPulse was built by a team of engineering technology students at El Sewedy University of Technology, combining full-stack web development, UX design, and market research to address a real gap in Egypt's pet care industry.

Name	Role	Contribution & Background
Salma Mohamed	frontend	Co-led the development of the PetPulse user interface using React.js and Vite. Salma focused on building out the core user flows, including the provider search interfaces and the adoption hub module. She ensured that the platform's UI components were highly responsive across all devices and implemented state management to handle complex user interactions seamlessly.
Laila Mohamed	backend	Contribution & Background: Architected and deployed the core backend infrastructure using Node.js and Express.js. Laila was responsible for designing the complex relational database schema in Supabase (PostgreSQL) to support heterogeneous data models for vets, trainers, and shops. She also developed the primary RESTful API endpoints and managed the seamless integration of Cloudinary for media storage.
Fatma Mohamed	backend	Contribution & Background: Focused on the backend logic and third-party integrations critical to the platform's functionality. Fatma engineered the booking logic, appointment scheduling systems, and the integration of the Paymob payment gateway. She also implemented database connection pooling and cron jobs to eliminate serverless cold-start latencies and maintain high availability.
Sara Salah	frontend	Co-led the frontend engineering effort with a strong focus on UI/UX design and styling using Tailwind CSS. Sara developed the reusable component library (cards, modals, form controls) used across the five main service modules. She was responsible for optimizing the frontend performance, ensuring the

Name	Role	Contribution & Background
		application met its target of sub-2-second page load times.
Ahmed Sherif	Agentic Ai	Spearheaded the research and integration of the OpenAI API to power PetPulse's intelligent features. Ahmed designed the system architecture required to support the Agentic AI Copilot, ensuring that user prompts were efficiently processed and contextually relevant responses were generated to assist pet owners with health and training queries.
Essam Osama	Agentic Ai	Focused on AI workflow automation and system integration. Essam developed the backend logic necessary to connect the AI models with the platform's live database, allowing the AI to fetch real-time data about provider availability and pet records. He optimized the API calls to reduce latency and manage token costs.
Gaber Hosny	security	Led the implementation of the platform's authentication and authorization architecture. Gaber integrated robust JWT (JSON Web Token) middleware to secure all write endpoints across the 19 Express routers, ensuring that stateless, horizontally scalable sessions were both reliable and highly secure against hijacking attempts.
Mohamed Ebrahim	Agentic Ai	Led the prompt engineering and AI model fine-tuning efforts. Mohamed established the strict safety guardrails and system instructions that prevent the AI from giving unverified medical advice, ensuring it properly redirects critical cases to human veterinarians within the PetPulse ecosystem.
Mahmoud Mustafa Mahmoud	security	Focused on data privacy and strict Role-Based Access Control (RBAC). Mahmoud engineered the logic that firmly separates the privileges of standard pet owners, verified service providers (vets/trainers), and platform administrators, ensuring that sensitive user data and financial records are strictly isolated
Abdelrahman Essam Ali	security	Directed the platform's vulnerability management and secure data transmission protocols. Abdelrahman ensured all database queries were sanitized to prevent SQL injection, managed the secure environment variables for production deployments, and validated the security compliance of the Paymob transaction processing flows.

This project was developed at El Sewedy University of Technology (SUT) — Egypt's polytechnic university dedicated to applied engineering and industry collaboration. SUT students work on projects connected to real industrial partners across the energy, manufacturing, and technology sectors.